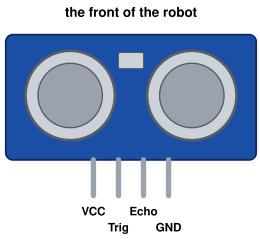
# How to plug in the distance sensor

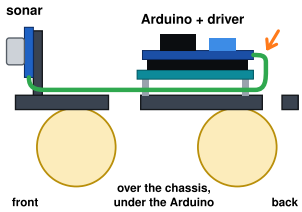
Then use it in Blockly: if it sees something, do this, else do that

## 1. Find the sonar



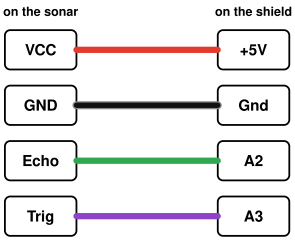

The **sonar** is the pair of round **"eyes"** at the front of the robot. Its **4 pins** are labeled **VCC**, **Trig**, **Echo**, **GND**.

## 2. Follow the wires



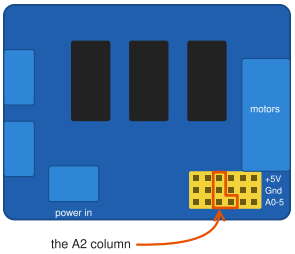

Look from the side. The four wires start **just behind the sonar**, run back **over the chassis**, **under the Arduino**, and loop up into the **driver board** at the back of the tower.

## 3. Match the wires



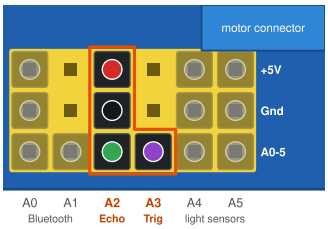

**VCC** goes to **+5V**, **GND** to **Gnd**, **Echo** to **A2**, **Trig** to **A3**. Follow each wire from the sonar back to the shield. Your wires can be any colors.

## 4. Find the analog header



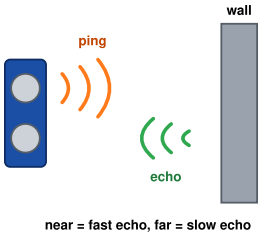

The other ends go on the motor shield, on the analog header: the **top row is all +5V**, the **middle row is all Gnd**, the **bottom row is A0 to A5**. The sonar uses the **third column, A2**.

## 5. Make an L



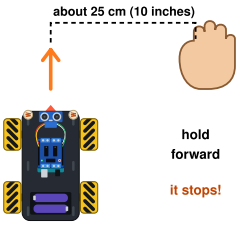

The four wires make an **L** shape.

## 6. Learn how it sees



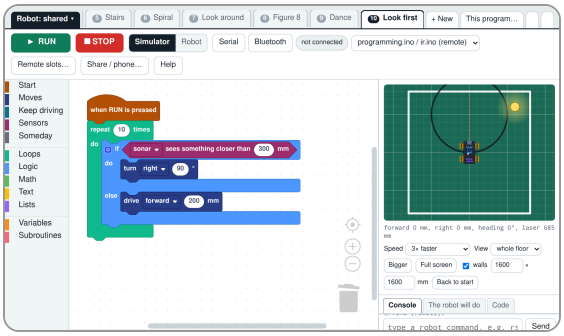

Trig sends a **ping** of sound too high to hear. Echo listens for it to bounce back: the longer the wait, the **farther away**. Soft or thin things are hard to see.

## 7. Check it




Hold **forward** on the remote and put your hand in front: the robot **stops about 25 cm away**. (It needs robot program **0.10.88** or newer.)

## 8. Program it: if ... else




Blockly

**robotics.makerlabkids.com/mecanum/blocks**

In **Sensors**, find **sonar sees something closer than 300 mm**. Put it in an **if ... else**: if it sees something, turn; else, go forward. Or press **+ New** and pick **Look first**. Connect with **Bluetooth**, choose **Robot**, press **RUN**. Programs with **if** run from the computer, not from a remote slot.